K–whole number a decimal or a fraction needed for making back pitch as an odd number.

Coil pitch, Y = ± 2m
m = 1 for simplex winding
m = 2 for duplex winding

Here '+' symbols denotes the progressive winding while '-' sign denotes the retrogressive winding. Generally all machines are winded by progressive method and hence we are using this type of winding methods in this lesson.

Front pitch, $Y_f = Y_b - Y$

v. Formula for wave winding

$$\text{Back pitch, } Y_b = \frac{2C}{P} \pm K$$

Here also 'C' denotes the number of coils and 'P' denotes the number of poles and the 'K' may be a whole number or a decimal or a fraction needed for making back pitch as an odd number.

$$\text{Coil pitch, } Y = \frac{2C \pm 2m}{P/2}$$

Here, m = 1 (for simplex)

m = 2 (for duplex)

Front pitch, $Y_f = Y - Y_b$

In wave winding, back pitch and front pitch must be equal.

Hence, $Y_b = Y_f$

vi. In double layer windings, the top layer coils should be numbered by odd numbers and the bottom layer should be numbered by even numbers.

vii. To find the current directions, use Fleming's right hand rule for generators and use Fleming's left hand rule for motors.

viii. The number of brushes must be equal to the number of parallel paths. For lap winding, the number of parallel paths will be equal to the number of poles and for wave winding, the number of parallel paths should be equal to 2

Hence, for lap winding,

Number of brushes = number of poles

For wave winding,
Number of brushes = 2

ix. Mark positive brush connected lead as 'A' and negative brush connected lead as "AA"
These two leads will be the armature leads of the D.C machine.

x. Draw solid vertical lines of equal length for the top layer coil sides and draw dotted vertical lines of same length for the bottom layer coil sides.

9.5.11 Double layer simplex Lap winding diagram

Draw the double layer simplex lap winding diagram for a 2 poles, 6 slots DC machine having 6 commutator segments. Indicate the position of brushes.

Solution

Number of poles P = 2
Number of slots S = 6
Number of commutator segments = 6

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